Illustration by **SIDDHANT JUMDE**



THE GIANT STEPS

- **India has only modest quantities of uranium, the only naturally available resource for obtaining fission energy. This has necessitated our thrust towards converting thorium into uranium–233**
- **To meet the urgent need for clean energy, quicker introduction of thorium in pressurised heavy water reactors (PHWRs) would address the potentially unmanageable issue of large spent fuel inventory arising from the use of natural uranium**
- **It would make PHWRs an attractive proposition for emerging economies countries**

We are rightly engaged in rapid renewable energy development. However, the total renewable energy potential in the country, including large hydro, is not likely to exceed 8,000 TWh/yr. This quantum of energy may be sufficient to meet today's energy needs. But how do we get two and a half times additional clean energy needed for Viksit Bharat? Clearly, thorium is the only answer. While we are progressing with accelerated deployment of PHWRs and even imported LWRs (light water reactors) to meet urgent clean energy needs, quicker introduction of thorium in PHWRs would address the potentially unmanageable issue of large spent fuel inventory arising as a result of the use of natural uranium on one side and make our PHWRs an attractive proposition for emerging economy countries on the other. In terms of economics and performance, they already are very competitive. Leveraging thorium thus helps address both domestic needs and support our export ambition. Since we must master the fuel recycle technologies of stage two and three in any case, this approach is also consistent with our planned three-stage programme.

Leveraging civil nuclear cooperation: We must also recognise the importance of leveraging our international civil nuclear cooperation, particularly with the United States, in realising such twin objectives. Nuclear fuel supply eventually becomes the larger business as compared to supply of nuclear reactors. High assay low enriched uranium (HALEU) is fast becoming an attractive proposition worldwide for various emerging nuclear reactor technologies. There is thus an urgent need for early demand signal for HALEU production infrastructure to grow. Deployment of thorium in PHWRs as highlighted earlier is precisely aligned to such an objective and can be deployed quickly through collaborative efforts between the two countries. As we approach net zero, the rising demand for clean energy would put global uranium supplies under considerable strain. India is thus well positioned to leverage her thorium resources and PHWR expertise to address the energy security of India and the world. ■

The author is former Chairman, Atomic Energy Commission